
1. Calculator (Sum, Subtraction, Multiplication, Division, Modulus, Average)

```
#include <stdio.h>

int main() {
    float num1, num2;

    printf("Enter two numbers: ");
    scanf("%f %f", &num1, &num2);

    printf("Sum: %.2f\n", num1 + num2);
    printf("Subtraction: %.2f\n", num1 - num2);
    printf("Multiplication: %.2f\n", num1 * num2);

    if(num2 != 0) {
        printf("Division: %.2f\n", num1 / num2);
        printf("Modulus: %d\n", (int)num1 % (int)num2);
    } else {
        printf("Cannot divide by zero\n");
    }

    printf("Average: %.2f\n", (num1 + num2) / 2);

    return 0;
}
```

2. Swap Two Numbers

```
#include <stdio.h>

int main() {
    int a, b, temp;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("Before swapping: a = %d, b = %d\n", a, b);

    temp = a;
    a = b;
    b = temp;

    printf("After swapping: a = %d, b = %d\n", a, b);

    return 0;
}
```

3. Swap Using XOR Bitwise Operator

```
#include <stdio.h>

int main() {
    int a, b;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("Before swapping: a = %d, b = %d\n", a, b);

    a = a ^ b;
    b = a ^ b;
    a = a ^ b;

    printf("After swapping: a = %d, b = %d\n", a, b);

    return 0;
}
```

4. Convert Days to Weeks and Days

```
#include <stdio.h>

int main() {
    int days, weeks, remainingDays;

    printf("Enter number of days: ");
    scanf("%d", &days);

    weeks = days / 7;
    remainingDays = days % 7;

    printf("%d days = %d weeks and %d days\n", days, weeks, remainingDays);

    return 0;
}
```

5. Rectangle Perimeter

```
#include <stdio.h>

int main() {
    float length, breadth, perimeter;

    printf("Enter length and breadth: ");
    scanf("%f %f", &length, &breadth);

    perimeter = 2 * (length + breadth);
}
```

```
    printf("Perimeter of rectangle: %.2f\n", perimeter);

    return 0;
}
```

6. Circle - Diameter, Circumference, and Area

```
#include <stdio.h>

int main() {
    float radius, diameter, circumference, area;
    const float PI = 3.14159;

    printf("Enter radius of circle: ");
    scanf("%f", &radius);

    diameter = 2 * radius;
    circumference = 2 * PI * radius;
    area = PI * radius * radius;

    printf("Diameter: %.2f\n", diameter);
    printf("Circumference: %.2f\n", circumference);
    printf("Area: %.2f\n", area);

    return 0;
}
```

7. Celsius to Fahrenheit

```
#include <stdio.h>

int main() {
    float celsius, fahrenheit;

    printf("Enter temperature in Celsius: ");
    scanf("%f", &celsius);

    fahrenheit = (celsius * 9/5) + 32;

    printf("%.2f°C = %.2f°F\n", celsius, fahrenheit);

    return 0;
}
```

8. Find Third Angle of Triangle

```
#include <stdio.h>

int main() {
```

```
float angle1, angle2, angle3;

printf("Enter two angles: ");
scanf("%f %f", &angle1, &angle2);

angle3 = 180 - (angle1 + angle2);

printf("Third angle: %.2f degrees\n", angle3);

return 0;
}
```

9. Area of Equilateral Triangle

```
#include <stdio.h>
#include <math.h>

int main() {
    float side, area;

    printf("Enter side of equilateral triangle: ");
    scanf("%f", &side);

    area = (sqrt(3) / 4) * side * side;

    printf("Area of equilateral triangle: %.2f\n", area);

    return 0;
}
```

10. Square Root of a Number

```
#include <stdio.h>
#include <math.h>

int main() {
    float num, result;

    printf("Enter a number: ");
    scanf("%f", &num);

    if(num >= 0) {
        result = sqrt(num);
        printf("Square root of %.2f = %.2f\n", num, result);
    } else {
        printf("Cannot find square root of negative number\n");
    }

    return 0;
}
```

11. Largest of Three Numbers

```
#include <stdio.h>

int main() {
    int a, b, c, largest;

    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);

    // Method 1: Using if-else
    if(a >= b && a >= c) {
        largest = a;
    } else if(b >= a && b >= c) {
        largest = b;
    } else {
        largest = c;
    }
    printf("Largest number (if-else): %d\n", largest);

    // Method 2: Using conditional operator
    largest = (a > b) ? ((a > c) ? a : c) : ((b > c) ? b : c);
    printf("Largest number (ternary): %d\n", largest);

    return 0;
}
```

12. Even or Odd

```
#include <stdio.h>

int main() {
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    // Method 1: Using if-else
    if(num % 2 == 0) {
        printf("%d is Even (if-else)\n", num);
    } else {
        printf("%d is Odd (if-else)\n", num);
    }

    // Method 2: Using conditional operator
    printf("%d is %s (ternary)\n", num, (num % 2 == 0) ? "Even" : "Odd");

    return 0;
}
```

13. Check if Character is Alphabet

```
#include <stdio.h>

int main() {
    char ch;

    printf("Enter a character: ");
    scanf(" %c", &ch);

    if((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z')) {
        printf("'%c' is an alphabet\n", ch);
    } else {
        printf("'%c' is not an alphabet\n", ch);
    }

    return 0;
}
```

14. Vowel or Consonant

```
#include <stdio.h>
#include <ctype.h>

int main() {
    char ch;

    printf("Enter an alphabet: ");
    scanf(" %c", &ch);

    ch = tolower(ch);

    if(ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
        printf("'%c' is a Vowel\n", ch);
    } else if((ch >= 'a' && ch <= 'z')) {
        printf("'%c' is a Consonant\n", ch);
    } else {
        printf("'%c' is not an alphabet\n", ch);
    }

    return 0;
}
```

15. Divisible by 5 and 11

```
#include <stdio.h>

int main() {
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if(num % 5 == 0 && num % 11 == 0) {
```

```

        printf("%d is divisible by both 5 and 11\n", num);
    } else {
        printf("%d is not divisible by both 5 and 11\n", num);
    }

    return 0;
}

```

16. Grade Calculator

```

#include <stdio.h>

int main() {
    int marks;
    char grade;

    printf("Enter marks (0-100): ");
    scanf("%d", &marks);

    // Method 1: Using if-else
    printf("\nMethod 1: Using if-else\n");
    if(marks >= 90) {
        grade = 'A';
    } else if(marks >= 80) {
        grade = 'B';
    } else if(marks >= 70) {
        grade = 'C';
    } else if(marks >= 60) {
        grade = 'D';
    } else if(marks >= 50) {
        grade = 'E';
    } else {
        grade = 'F';
    }
    printf("Grade: %c\n", grade);

    // Method 2: Using switch-case
    printf("\nMethod 2: Using switch-case\n");
    switch(marks / 10) {
        case 10:
        case 9:
            printf("Grade: A\n");
            break;
        case 8:
            printf("Grade: B\n");
            break;
        case 7:
            printf("Grade: C\n");
            break;
        case 6:
            printf("Grade: D\n");
            break;
        case 5:
            printf("Grade: E\n");

```

```

        break;
    default:
        printf("Grade: F\n");
    }

    return 0;
}

```

17. Factorial

```

#include <stdio.h>

int main() {
    int num, i;
    unsigned long long factorial = 1;

    printf("Enter a number: ");
    scanf("%d", &num);

    if(num < 0) {
        printf("Factorial not defined for negative numbers\n");
    } else {
        for(i = 1; i <= num; i++) {
            factorial *= i;
        }
        printf("Factorial of %d = %llu\n", num, factorial);
    }

    return 0;
}

```

18. Fibonacci Series

```

#include <stdio.h>

int main() {
    int n, i;
    unsigned long long first = 0, second = 1, next;

    printf("Enter number of terms: ");
    scanf("%d", &n);

    printf("Fibonacci Series: ");

    for(i = 0; i < n; i++) {
        if(i <= 1) {
            next = i;
        } else {
            next = first + second;
            first = second;
            second = next;
        }
    }
}

```



```

        printf("%llu ", next);
    }
    printf("\n");

    return 0;
}

```

19. Matrix Addition and Subtraction

```

#include <stdio.h>

int main() {
    int rows, cols, i, j;
    int matrix1[10][10], matrix2[10][10], sum[10][10], diff[10][10];

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &rows, &cols);

    printf("Enter elements of first matrix:\n");
    for(i = 0; i < rows; i++) {
        for(j = 0; j < cols; j++) {
            scanf("%d", &matrix1[i][j]);
        }
    }

    printf("Enter elements of second matrix:\n");
    for(i = 0; i < rows; i++) {
        for(j = 0; j < cols; j++) {
            scanf("%d", &matrix2[i][j]);
        }
    }

    // Addition and Subtraction
    for(i = 0; i < rows; i++) {
        for(j = 0; j < cols; j++) {
            sum[i][j] = matrix1[i][j] + matrix2[i][j];
            diff[i][j] = matrix1[i][j] - matrix2[i][j];
        }
    }

    printf("\nSum of matrices:\n");
    for(i = 0; i < rows; i++) {
        for(j = 0; j < cols; j++) {
            printf("%d ", sum[i][j]);
        }
        printf("\n");
    }

    printf("\nDifference of matrices:\n");
    for(i = 0; i < rows; i++) {
        for(j = 0; j < cols; j++) {
            printf("%d ", diff[i][j]);
        }
        printf("\n");
    }
}

```

```
    }

    return 0;
}
```

STRING PROGRAMS

1. String Length (without strlen)

```
#include <stdio.h>

int main() {
    char str[100];
    int length = 0;

    printf("Enter a string: ");
    scanf("%[^\\n]", str);

    while(str[length] != '\\0') {
        length++;
    }

    printf("Length of string: %d\\n", length);

    return 0;
}
```

2. Reverse String (without strrev)

```
#include <stdio.h>

int main() {
    char str[100], rev[100];
    int length = 0, i;

    printf("Enter a string: ");
    scanf("%[^\\n]", str);

    // Find length
    while(str[length] != '\\0') {
        length++;
    }

    // Reverse the string
    for(i = 0; i < length; i++) {
        rev[i] = str[length - 1 - i];
    }
    rev[length] = '\\0';

    printf("Original string: %s\\n", str);
    printf("Reversed string: %s\\n", rev);
}
```

```

    return 0;
}

```

3. Palindrome Check

```

#include <stdio.h>
#include <ctype.h>

int main() {
    char str[100];
    int length = 0, i, isPalindrome = 1;

    printf("Enter a string: ");
    scanf("%[^\n]", str);

    // Find length
    while(str[length] != '\0') {
        length++;
    }

    // Check palindrome
    for(i = 0; i < length / 2; i++) {
        if(tolower(str[i]) != tolower(str[length - 1 - i])) {
            isPalindrome = 0;
            break;
        }
    }

    if(isPalindrome) {
        printf("%s' is a palindrome\n", str);
    } else {
        printf("%s' is not a palindrome\n", str);
    }

    return 0;
}

```

4. Toggle Case

```

#include <stdio.h>

int main() {
    char str[100];
    int i;

    printf("Enter a string: ");
    scanf("%[^\n]", str);

    printf("Original string: %s\n", str);

    for(i = 0; str[i] != '\0'; i++) {

```

```
cook@pop-os:~$ ascii -d
```

0	NUL	16	DLE	32	48	0	64	@	80	P	96	`	112	p	
1	SOH	17	DC1	33	!	49	1	65	A	81	Q	97	a	113	q
2	STX	18	DC2	34	"	50	2	66	B	82	R	98	b	114	r
3	ETX	19	DC3	35	#	51	3	67	C	83	S	99	c	115	s
4	EOT	20	DC4	36	\$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
6	ACK	22	SYN	38	&	54	6	70	F	86	V	102	f	118	v
7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
8	BS	24	CAN	40	(	56	8	72	H	88	X	104	h	120	x
9	HT	25	EM	41	)	57	9	73	I	89	Y	105	i	121	y
10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z
11	VT	27	ESC	43	+	59	;	75	K	91	[	107	k	123	{
12	FF	28	FS	44	,	60	<	76	L	92	\	108	l	124	
13	CR	29	GS	45	-	61	=	77	M	93	]	109	m	125	}
14	SO	30	RS	46	.	62	>	78	N	94	^	110	n	126	~
15	SI	31	US	47	/	63	?	79	O	95	_	111	o	127	DEL

```

        if(str[i] >= 'a' && str[i] <= 'z') {
            str[i] = str[i] - 32; // Convert to uppercase
        } else if(str[i] >= 'A' && str[i] <= 'Z') {
            str[i] = str[i] + 32; // Convert to lowercase
        }
    }

    printf("Toggled string: %s\n", str);

    return 0;
}

```

SERIES AND FUNCTIONS

1. Series: 3, 4, 6, 9, 13, ...

```

#include <stdio.h>

int main() {
    int n, i, term = 3, diff = 1;

    printf("Enter which term to find: ");
    scanf("%d", &n);

    for(i = 1; i < n; i++) {
        term += diff;
        diff++;
    }

    printf("The %dth term is: %d\n", n, term);

    return 0;
}

```

2. Series: 1, 3, 7, 13, 21, ...

```

#include <stdio.h>

int main() {
    int n, i, term = 1, diff = 2;

    printf("Enter which term to find: ");
    scanf("%d", &n);

    for(i = 1; i < n; i++) {
        term += diff;
        diff += 2;
    }

    printf("The %dth term is: %d\n", n, term);
}

```

```
    return 0;
}
```

3. Even or Odd Using Function

```
#include <stdio.h>

int isEven(int num) {
    return (num % 2 == 0);
}

int main() {
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if(isEven(num)) {
        printf("%d is Even\n", num);
    } else {
        printf("%d is Odd\n", num);
    }

    return 0;
}
```

4. String Comparison

```
#include <stdio.h>

int main() {
    char str1[100], str2[100];
    int i = 0, result;

    printf("Enter first string: ");
    scanf("%s", str1);
    getchar(); // Clear buffer
    printf("Enter second string: ");
    scanf("%s", str2);

    // Manual comparison
    while(str1[i] != '\0' && str2[i] != '\0') {
        if(str1[i] != str2[i]) {
            break;
        }
        i++;
    }

    result = str1[i] - str2[i];

    if(result > 0) {
        printf("'s' is greater than '%s'\n", str1, str2);
    }
}
```

```
    } else if(result < 0) {
        printf("'%' is less than '%s'\n", str1, str2);
    } else {
        printf("Both strings are equal\n");
    }

    return 0;
}
```

5. Trapezium Area Using Function

```
#include <stdio.h>

float trapeziumArea(float a, float b, float h) {
    return ((a + b) / 2) * h;
}

int main() {
    float a, b, h, area;

    printf("Enter length of parallel side 1: ");
    scanf("%f", &a);
    printf("Enter length of parallel side 2: ");
    scanf("%f", &b);
    printf("Enter height: ");
    scanf("%f", &h);

    area = trapeziumArea(a, b, h);

    printf("Area of trapezium: %.2f\n", area);

    return 0;
}
```

6. Prime Number Check Using Function

```
#include <stdio.h>
#include <math.h>

int isPrime(int num) {
    int i;

    if(num <= 1) {
        return 0;
    }

    for(i = 2; i <= sqrt(num); i++) {
        if(num % i == 0) {
            return 0;
        }
    }
}
```

```
        return 1;
    }

int main() {
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if(isPrime(num)) {
        printf("%d is a Prime number\n", num);
    } else {
        printf("%d is not a Prime number\n", num);
    }

    return 0;
}
```
